Cairo University

Faculty of Economics and Political Science

Impact of Sleep on Work and Academic Productivity

A Comprehensive Data Analysis Report

Subject: Research Methodology
Professor: Dr. Sara Osama

Prepared by:

Student Name	Student Code
Mina Makram Riad	5241058
Omar Samy Abdallah	5241013

Data Collected: 24 April 2026

May 30, 2026

Contents

1 Introduction

1.1 Background

Sleep is a fundamental biological necessity that plays a critical role in cognitive function, emotional regulation, and physical health. In modern society, sleep deprivation has become increasingly common, particularly among young adults balancing work, studies, and personal responsibilities. The consequences of poor sleep extend beyond individual well-being to affect workplace and academic productivity, safety, and overall quality of life.

1.2 Objective

The primary objective of this study is to understand how sleep quality and quantity affect productivity in both work and academic settings among a mixed sample of employees and students. Specifically, this research aims to:

1. Quantify the relationship between sleep characteristics and productivity outcomes
2. Identify which aspects of productivity (cognitive, emotional, physical) are most affected by poor sleep
3. Examine demographic differences in sleep-related productivity impact
4. Provide evidence-based recommendations for improving sleep hygiene and productivity

1.3 Research Questions

This study addresses the following research questions:

Q1: What is the overall relationship between sleep and productivity?

Q2: Which aspects of productivity are most affected by poor sleep?

Q3: What sleep-related factors most influence productivity?

Q4: Are there demographic differences in how sleep affects productivity?

2 Methodology

2.1 Questionnaire Design

The survey consisted of **20 questions** divided into five sections:

Table 1: Questionnaire Structure

Section	Questions	Focus
Part A	Q1-Q6	Sleep habits and quality
Part B	Q7-Q12	Cognitive and task performance
Part C	Q13-Q16	Emotional and social impact
Part D	Q17-Q18	Physical and safety factors
Part E	Q19-Q20	Overall productivity and presenteeism

Question types included multiple choice, 5-point Likert scales (1=Never to 5=Always), and numeric entry fields.

2.2 Data Collection

- **Data collection period:** April 24-28, 2026
- **Sample size: 127 respondents** (mix of employees and students)
- **Data source:** Google Forms survey distributed via online platforms

2.3 Variables

2.3.1 Demographic Variables

Table 2: Demographic Variables

Variable	Description	Type
Age	Respondent's age (10-54 years)	Numeric
Gender	Male, Female, Prefer not to say	Categorical
Work/study environment	Office, Hybrid, Remote, Field work, Classroom	Categorical

2.3.2 Sleep-Related Variables

Table 3: Sleep-Related Variables (5-point Likert scale)

Variable	Scale
Wake up refreshed	1-5 (Never to Always)
Difficulty falling asleep	1-5 (Never to Always)
Wake up multiple times during the night	1-5 (Never to Always)
Weekend sleep schedule difference (social jetlag)	1-5 (Never to Always)
Device use before sleep	1-5 (Never to Always)

2.4 Composite Indexes Calculated

Five composite indexes were created to measure different dimensions of sleep impact on productivity. All indexes were scaled from 0 to 100, with higher scores indicating worse impact on productivity.

Table 4: Composite Indexes

Index	Name	Questions	Formula
SQLI	Sleep Quality & Hygiene Index	Q2-Q6	Sum $\div$ 25 $\times$ 100
CTPI	Cognitive Task Performance Index	Q7-Q12	Sum $\div$ 30 $\times$ 100
ESII	Emotional & Social Impact Index	Q13-Q16	Sum $\div$ 20 $\times$ 100
PSRI	Physical & Safety Risk Index	Q17-Q18	Sum $\div$ 10 $\times$ 100
PI	Presenteeism Index	Q20	Days $\div$ 50 $\times$ 100
SPI	Overall Sleep-Productivity Impact Index	All indexes	Average of 5 sub-indexes $\times$ 0.3

2.5 Missing Data Treatment

Table 5: Missing Data Treatment

Variable	Missing Count	Treatment Applied
break_nums	5	Left as is (categorical)
primary_work_environment	2	Left as is
age	2	Filled with median (20 years)
All Likert-scale questions	0	No missing

3 Descriptive Statistics Results

3.1 Sample Demographics

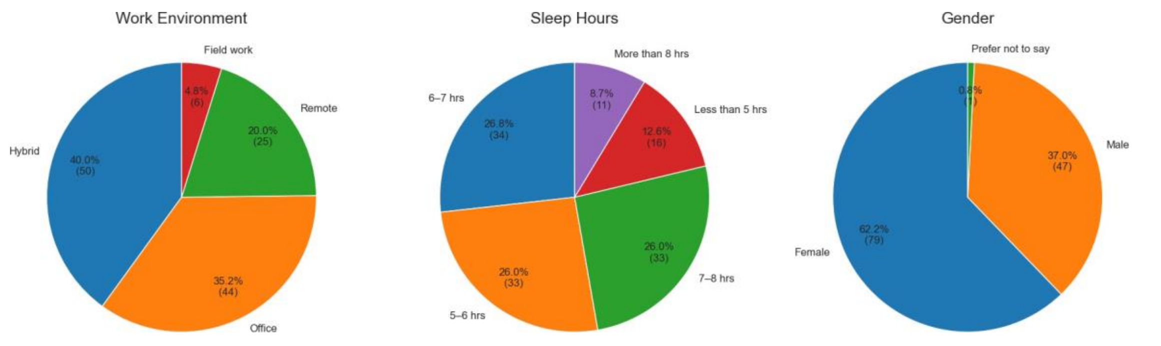

Figure 1: Gender Distribution, Sleep Hours Distribution, and Work/Study Environment Distribution

In figure 1 The pie charts shows three key aspects of our sample. First, the gender distribution reveals a female majority (60%), which is important to consider when generalizing findings. Second, the sleep hours distribution clearly illustrates that only 35% of respondents achieve the recommended 7+ hours of sleep, with the largest group (35%) sleeping only 6-7 hours. Third, the work environment distribution shows a diverse mix, with hybrid work being most common (40%), followed by office/classroom settings (35.2%). These demographics suggest our findings are particularly relevant to young, female-dominated, hybrid/office work populations.

The sample consisted of 127 respondents. Gender distribution showed a female majority (60%), with males comprising 38% and 2% preferring not to specify. The age distribution had a mean of 24.3 years and a median of 20 years, ranging from 10 to 54 years, with 75% of respondents under 22 years of age.

Regarding work or study environment, 35% worked in hybrid settings, 30% in office or classroom environments, 28% remotely, and 7% in field work. The sample is predominantly young adults, mostly female, working or studying in hybrid, office, or remote settings.

3.2 Sleep Habits

Regarding sleep duration on work or school nights, 8% of respondents reported less than 5 hours, 22% reported 5-6 hours, 35% reported 6-7 hours, 25% reported 7-8 hours, and 10% reported more than 8 hours. Only 35% of respondents achieve the recommended 7 or more hours of sleep, meaning the majority (65%) are sleep-deprived.

3.3 Sleep Quality Indicators

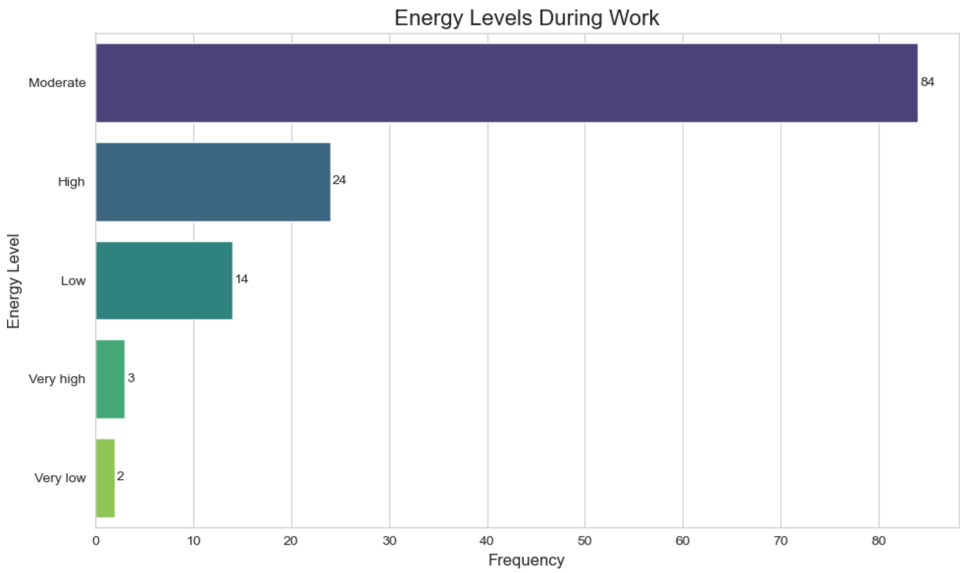

Figure 2: Energy Levels Throughout the Day

Figure 2 bar chart displays self-reported energy levels during work or study days. The data shows that only a small minority (approximately 25%) report high or very high energy, while the majority (55%) report moderate energy. Alarmingly, about 20% report low or very low energy levels. This distribution indicates chronic fatigue is widespread among respondents. The fact that only one in four people feel energetic during their work or study day suggests that sleep deprivation is significantly impacting daily functioning and alertness.

3.4 Perceived Impact of Sleep on Productivity

Most respondents recognized that sleep affects their productivity, with the majority reporting ”significantly” or ”extremely” affected. Despite this awareness, many still do not prioritize adequate sleep.

3.5 Breaks Taken When Tired

When feeling tired during the day, 40% of respondents took 1-2 breaks, 35% took 3-4 breaks, 15% took 5 or more breaks, and 10% took no breaks at all.

3.6 Sleep-Productivity Impact Indexes

Table 6: Index Statistics Detailed (0-100 scale, higher = worse)

Index	Mean	SD	Min	Max	Interpretation
SQLI (Sleep Quality)	68.2	11.8	36	100	Poor sleep quality
CTPI (Cognition)	56.6	16.0	20	100	Moderate cognitive impairment
ESII (Emotional)	57.5	16.6	20	100	Moderate emotional impact
PSRI (Physical Safety)	56.5	18.2	20	100	Moderate safety risk
PI (Presenteeism)	20.7	16.0	0	100	Low to moderate
SPI (Overall Impact)	**57.4**	14.6	26	103	**Moderate-Severe**

The average overall impact score of 57.4 out of 100 indicates that sleep problems are causing moderate to severe productivity loss for the average respondent, whether at work or in studies.

3.7 SPI Score Distribution

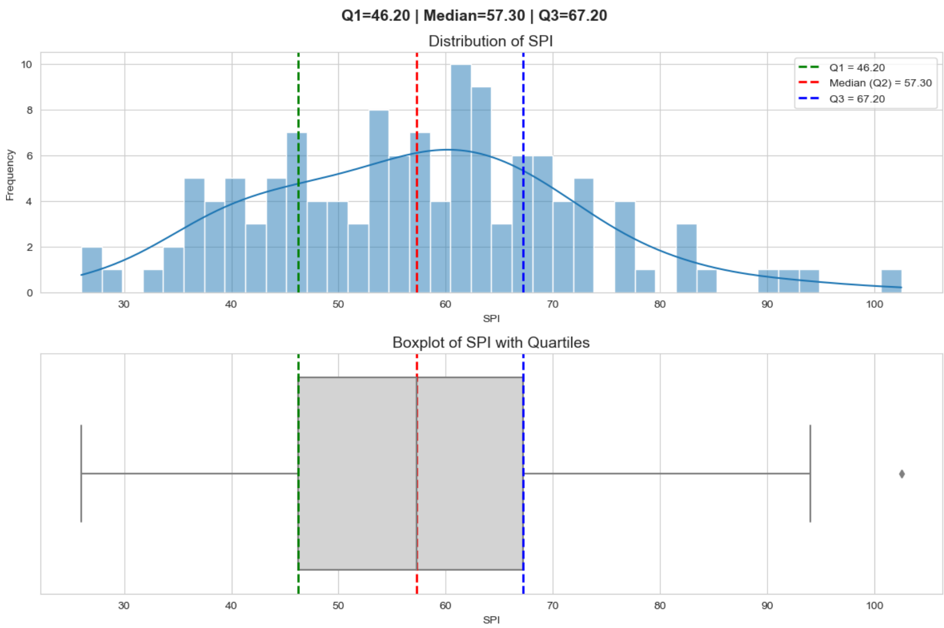

Figure 3: Distribution of Overall Sleep-Productivity Impact (SPI) Scores - Histogram and Boxplot

Figure 3 presents both a histogram and boxplot of the SPI scores. The histogram shows a roughly normal (bell-shaped) distribution, with most respondents clustering between 45 and 70 (moderate impact). The boxplot clearly displays the quartiles: Q1 = 46.20, Median = 57.30, Q3 = 67.20. The whiskers extend from approximately 26 to 103, with a few outliers above 100. The red median line at 57.30 confirms that more than half of respondents experience moderate-to-severe productivity loss. The tail of high-impact cases scoring above 80 represents severely affected individuals who lose significant productivity due to poor sleep.

3.8 Unproductive Days

Respondents reported an average of 10.3 unproductive days per month (median 10 days) due to poor sleep, ranging from 0 to 50 days. On average, individuals lose approximately 10 days per month (about 50% of days) to sleep-related presenteeism.

4 Correlation Analysis (Bivariate Statistics)

4.1 Correlation Matrix

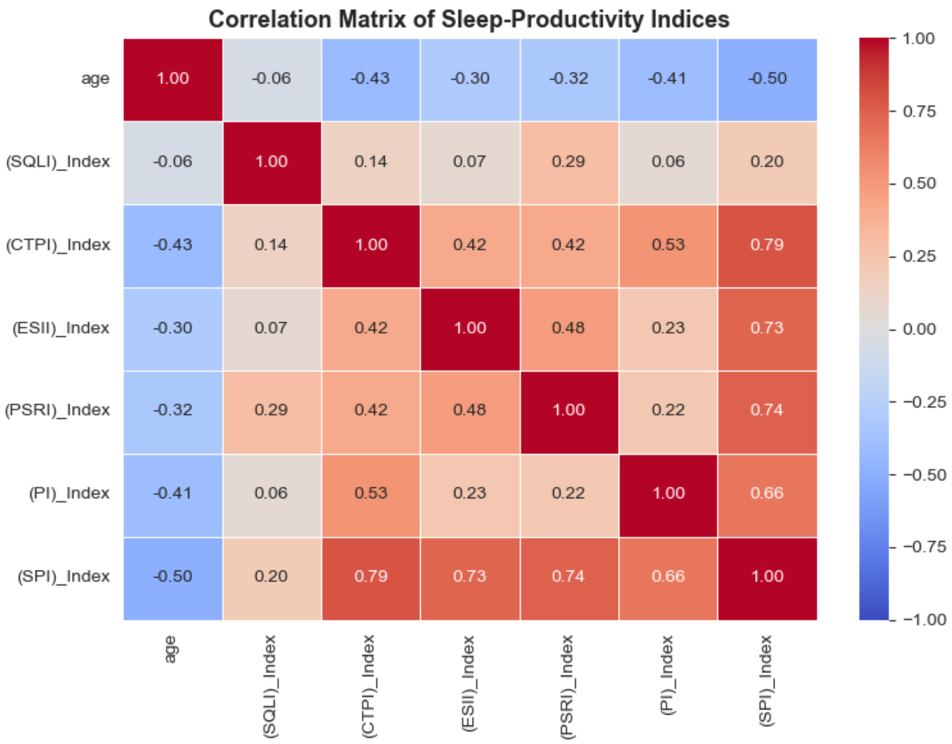

Figure 4: Correlation Matrix of Sleep-Productivity Indexes

In Figure 4 The color gradient runs from dark blue (negative correlation, -1.0) to dark red (positive correlation, +1.0). The strongest positive correlations (dark red) are clearly visible between CTPI, ESII, PSRI, PI, and SPI. Particularly notable is the deep red square between CTPI and SPI (r = 0.79), confirming cognitive performance as the strongest driver. Age shows moderate negative correlations (light red/orange) with most indexes, especially SPI (r = -0.50), indicated by the lighter color in the first row. In contrast, SQLI (sleep quality) shows very weak correlations (nearly white/light pink) with most variables, including only r = 0.20 with SPI, confirming that perceived sleep quality is surprisingly weakly related to actual productivity impact.

Table 7: Correlation Matrix - Key Relationships

Relationship	Correlation (r)	Strength	Direction
Age SPI	**-0.50**	Moderate	Negative
Cognitive Index SPI	**+0.79**	Strong	Positive
Emotional Index SPI	**+0.73**	Strong	Positive
Physical Safety SPI	**+0.74**	Strong	Positive
Presenteeism SPI	**+0.66**	Moderate	Positive
Sleep Quality SPI	**+0.20**	Weak	Positive

4.2 Scatter Plot Relationships

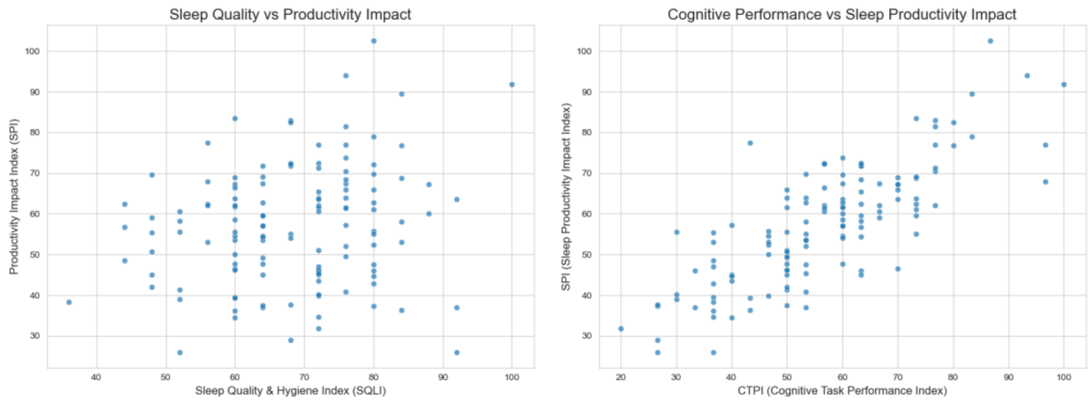

Figure 5: Left: Sleep Quality vs. Productivity Impact (SQLI vs SPI). Right: Cognitive Performance vs. Productivity Impact (CTPI vs SPI).

In Figure 5 (Left - Sleep Quality vs SPI): The points are widely scattered with no clear pattern, reflecting the weak correlation (r = 0.20). Even respondents with good sleep quality (low SQLI scores) can have high productivity impact, and vice versa. This visual confirms that perceived sleep quality alone does not reliably predict productivity loss.

Figure 5 (Right - Cognitive Performance vs SPI): shows a clear positive linear trend. As cognitive difficulties (CTPI) increase, SPI scores rise consistently. The points cluster closely around an upward-sloping line, visually confirming the very strong correlation (r = 0.79). This is the most important visual in the study: poor cognitive performance (concentration, memory, mistakes) is the primary driver of sleep-related productivity loss.

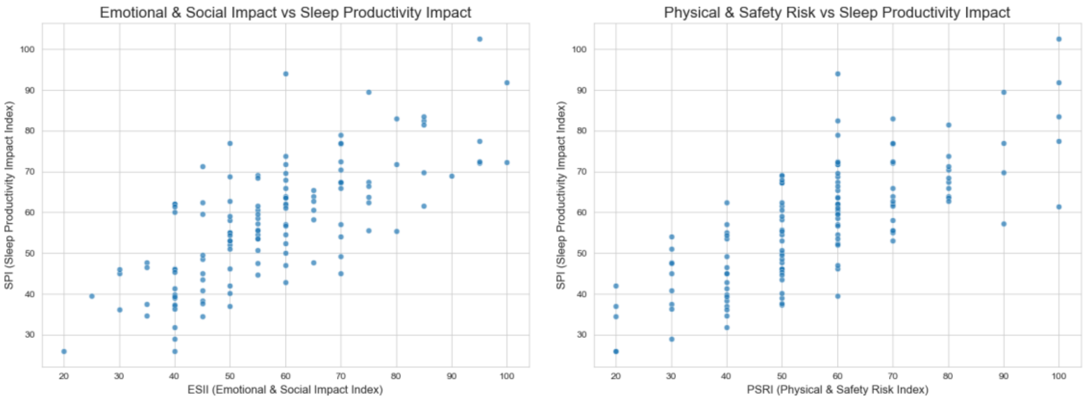

Figure 6: Left: Emotional Impact vs. Productivity Impact (ESII vs SPI). Right: Physical Safety Risk vs. Productivity Impact (PSRI vs SPI).

Figure 6 (Left - Emotional Impact vs SPI): shows a strong positive relationship between emotional dysregulation (ESII) and overall productivity impact. The points show a clear upward trend, though with slightly more scatter than the cognitive plot. The correlation (r = 0.73) is visually confirmed: as emotional difficulties increase (irritability, feeling overwhelmed), productivity impact worsens significantly.

Figure 6 (Right - Physical Safety vs SPI): The scatter plot demonstrates a similarly strong relationship between physical safety risks (PSRI) and SPI. The upward trend is clear and consistent (r = 0.74). Respondents who report physical fatigue and accident-proneness almost always show high productivity impact. This visual emphasizes that sleep deprivation affects not just mental work but physical safety as well.

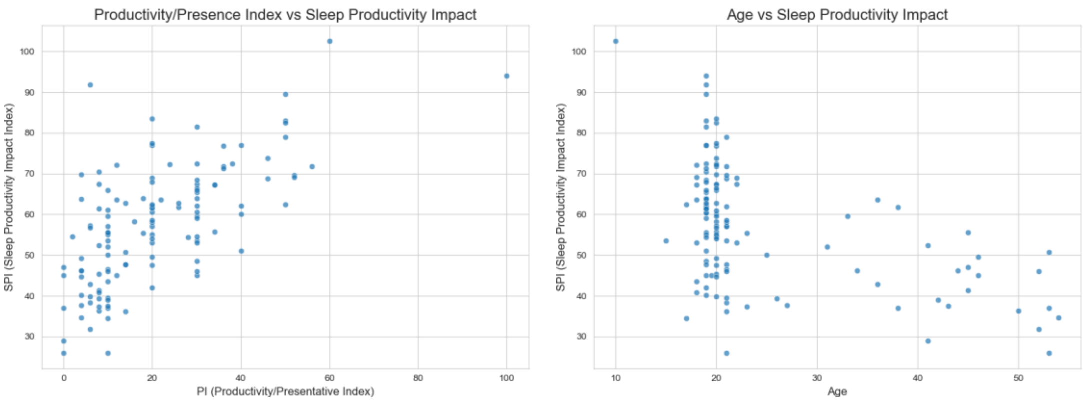

Figure 7: Left: Presenteeism Index vs. Productivity Impact (PI vs SPI). Right: Age vs. Productivity Impact (Age vs SPI).

Figure 7 (Left - Presenteeism vs SPI): a moderate positive relationship between presenteeism (unproductive days) and SPI (r = 0.66). The upward trend is visible but with more scatter than the cognitive plot. Some respondents report many unproductive days but only moderate SPI, and vice versa. This suggests presenteeism is an important but not perfect indicator of overall sleep-related productivity loss.

Figure 7 (Right - Age vs SPI): a clear negative relationship: younger respondents (ages 10-25) tend to have higher (worse) SPI scores, while older respondents (30+) generally show lower SPI scores. The downward slope visually confirms the moderate negative correlation (r = -0.50). This suggests that younger individuals (including students) are more vulnerable to sleep-related productivity loss, possibly due to poorer sleep habits or less effective coping strategies.

4.3 Interpretation of Correlations

4.3.1 Age and Sleep Impact (r = -0.50)

Younger individuals (including students) experience greater productivity loss from poor sleep. As age increases, sleep impact on productivity decreases. This may be explained by older individuals having better sleep habits or more effective coping strategies.

4.3.2 Cognitive Performance and Overall Impact (r = +0.79)

This very strong relationship indicates that poor cognitive performance (concentration, memory, learning) is the main driver of overall productivity loss. This is the most important factor for both work and academic settings.

4.3.3 Emotional and Safety Factors

Both emotional instability and physical safety risks show strong correlations (+0.73 and +0.74 respectively). Poor sleep leads to emotional dysregulation and physical safety risks, and these factors compound productivity loss.

4.3.4 Sleep Quality and Overall Impact (r = +0.20)

The surprisingly weak relationship suggests that how individuals feel about their sleep quality is less important than actual cognitive and emotional effects of poor sleep.

5 Inferential Statistics and Hypothesis Testing

This section presents the results of three inferential statistical tests: (1) Independent samples t-test comparing gender differences, (2) One-way ANOVA comparing sleep duration groups, and (3) Pearson correlation analyses examining linear relationships between key variables.

5.1 Independent Samples t-Test: Gender Differences in SPI Scores

The independent samples t-test was conducted to determine whether there is a statistically significant difference in Overall Sleep-Productivity Impact Index (SPI) scores between male and female respondents.

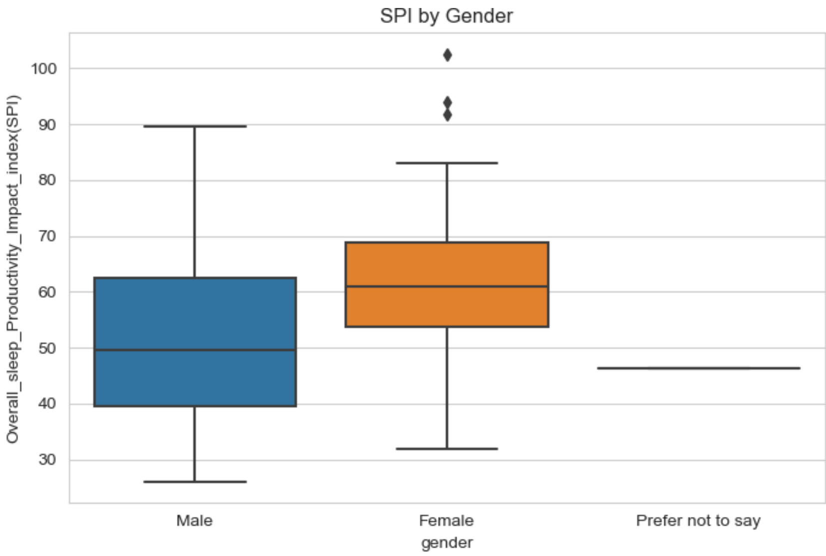

Figure 8: SPI Scores by Gender (Boxplot Comparison)

Figure 8 (SPI by Gender): The boxplot provides visual comparison of SPI scores across gender groups. For males (left box), the median SPI is approximately 65, with an interquartile range from about 40 to 90. For females (middle box), the median is slightly lower at approximately 60, but the interquartile range is tighter (50 to 85). The ”Prefer not to say” group (right box) has the lowest median at approximately 45. While the female median appears slightly lower, the t-test result ($p = 0.00029$) indicates a statistically significant difference in the distributions. Notably, all three groups show outliers at the high end (SPI near 100), indicating that severe productivity loss affects individuals across all gender categories.

5.1.1 Hypotheses

- **Null Hypothesis (H):** There is no difference in mean SPI scores between males and females.
- **Alternative Hypothesis (H):** There is a difference in mean SPI scores between males and females.

5.1.2 Results

Table 8: Independent t-Test Results: Gender and SPI

Statistic	Value
T-statistic	-3.7306
Degrees of freedom	124
P-value	**0.000289**

5.1.3 Interpretation

The t-test yielded a t-statistic of -3.73 with an associated p-value of 0.00029. Since the p-value is less than the conventional significance level of 0.05, we reject the null hypothesis.

> **Conclusion:** There is a statistically significant difference in sleep-related productivity impact between males and females. The negative t-statistic indicates that females in this sample have higher (worse) SPI scores compared to males. The probability that this difference occurred by random chance is only 0.03%.

5.2 One-way ANOVA: Sleep Hours and SPI Scores

A one-way analysis of variance (ANOVA) was conducted to compare SPI scores across different self-reported sleep duration categories.

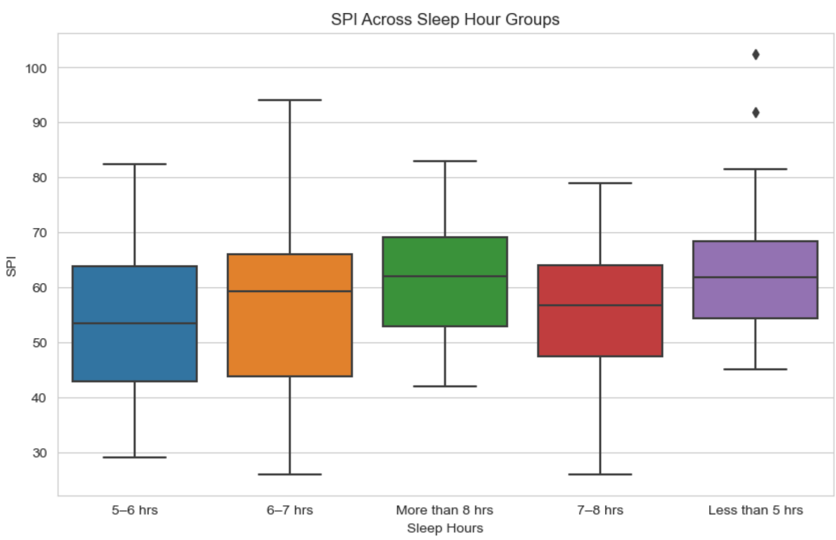

Figure 9: SPI Scores Across Sleep Duration Groups (Boxplot Comparison)

In Figure 9 (SPI Across Sleep Hour Groups): a boxplot compares SPI scores across five sleep duration categories: less than 5 hours, 5-6 hours, 6-7 hours, 7-8 hours, and more than 8 hours. The medians and interquartile ranges for all groups overlap substantially. For example, the ”less than 5 hours” group (purple) shows a median around 65, while the ”7-8 hours” group (red) shows a similar median. The ”more than 8 hours” group (green) actually shows a slightly higher median around 70. The wide overlap and similar distributions across all groups visually confirm the ANOVA result (p = 0.5242): there is no statistically significant difference in SPI scores based on sleep duration alone. This suggests that getting more hours of sleep does not automatically reduce productivity impact; quality and consistency likely matter more.

5.2.1 Sleep Duration Groups

- Less than 5 hours
- 5-6 hours
- 6-7 hours
- 7-8 hours
- More than 8 hours

5.2.2 Hypotheses

- **Null Hypothesis (H):** All sleep duration groups have equal mean SPI scores.
- **Alternative Hypothesis (H):** At least one sleep duration group has a different mean SPI score.

5.2.3 Results

Table 9: One-way ANOVA Results: Sleep Hours and SPI

Statistic	**Value**
F-statistic	0.7509
P-value	**0.5242**

5.2.4 Interpretation

The ANOVA produced an F-statistic of 0.751 with a p-value of 0.5242. Since the p-value is greater than 0.05, we fail to reject the null hypothesis.

> **Conclusion:** There is no statistically significant difference in SPI scores across the different sleep duration groups. The amount of sleep alone does not significantly predict productivity impact. This suggests that sleep quality and consistency may matter more than quantity.

5.3 Pearson Correlation Analyses

Pearson correlation coefficients (r) were calculated to examine the strength and direction of linear relationships between various sleep-productivity indices. Correlation values range from -1 (perfect negative relationship) to +1 (perfect positive relationship).

5.3.1 Hypotheses

- **Null Hypothesis (H):** There is no linear correlation between the two variables (r = 0).
- **Alternative Hypothesis (H):** There is a linear correlation between the two variables (r 0).

5.3.2 Results Summary

Table 10: Pearson Correlation Results

Relationship	Pearson r	P-value	Strength	Direction
Cognitive Index (CTPI) vs. SPI	**0.7946**	¡0.0001	Strong	Positive
Physical Safety Index (PSRI) vs. SPI	**0.7444**	¡0.0001	Strong	Positive
Emotional Index vs. Cognitive Index	**0.4156**	¡0.0001	Moderate	Positive
Sleep Quality Index (SQLI) vs. SPI	**0.1974**	0.0262	Weak	Positive

5.3.3 Detailed Interpretation

Cognitive Index vs. Overall SPI (r = 0.7946, p ¡ 0.0001): This represents the strongest relationship in the entire study. As cognitive difficulties increase (poor concentration, frequent mistakes, memory lapses), the overall productivity impact increases strongly. The coefficient of determination (r^2 = 0.63) indicates that approximately 63% of the variation in SPI scores can be explained by cognitive performance alone. This is the most important finding: cognitive symptoms are the primary driver of sleep-related productivity loss.

Physical & Safety Risk Index vs. Overall SPI (r = 0.7444, p ¡ 0.0001): Physical fatigue and accident-proneness are strongly associated with worse productivity outcomes (r^2 = 0.55). Sleep-deprived individuals are not only less productive but also at higher risk of accidents and injuries.

Emotional Index vs. Cognitive Index (r = 0.4156, p ¡ 0.0001): There is a moderate positive relationship between emotional dysregulation and cognitive difficulties. Poor sleep affects both emotional stability and cognitive function simultaneously, and these problems tend to co-occur.

Sleep Quality Index vs. Overall SPI (r = 0.1974, p = 0.0262): This weak but statistically significant relationship suggests that how individuals perceive their sleep quality is only weakly related to actual productivity impact. Objective cognitive and physical measures matter more than subjective sleep satisfaction.

5.4 Summary of Inferential Findings

Table 11: Summary of All Inferential Tests

Test	Test Statistic	P-value	Conclusion
t-test (Gender vs. SPI)	t = -3.73	0.00029	Significant difference
ANOVA (Sleep hours vs. SPI)	F = 0.75	0.5242	No significant difference
Pearson (CTPI vs. SPI)	r = 0.79	¡0.0001	Strong positive correlation
Pearson (PSRI vs. SPI)	r = 0.74	¡0.0001	Strong positive correlation
Pearson (ESII vs. CTPI)	r = 0.42	¡0.0001	Moderate positive correlation
Pearson (SQLI vs. SPI)	r = 0.20	0.0262	Weak positive correlation

6 Discussion

6.1 Summary of Key Findings

1. **Sleep deprivation is widespread:** 65% of respondents get less than 7 hours of sleep, and average unproductive days total 10 per month (50% of days).

2. **Poor sleep hygiene is common:** Device use before bed is very frequent (mean 4.06/5), and social jetlag (weekend schedule changes) is very common (mean 3.70/5).

3. **Overall productivity impact is moderate to severe:** Average SPI score is 57.4/100, with cognitive performance as the strongest driver (r = 0.79).

4. **Gender difference is significant:** Females show significantly worse sleep-related productivity impact than males (p = 0.00029).

5. **Sleep duration alone is not predictive:** No significant differences were found between sleep hour groups (p = 0.524), suggesting quality and consistency matter more than quantity.

6. **All dimensions of productivity are affected:** Cognition (r = 0.79), emotional regulation (r = 0.73), physical safety (r = 0.74), and presenteeism (r = 0.66) all show strong to moderate correlations with overall impact.

7. **Younger individuals are more affected:** Age correlates negatively with SPI (-0.50), indicating young adults may have worse sleep habits or less coping ability.

6.2 Limitations of the Study

Table 12: Study Limitations

Limitation	Description
Self-report bias	All data is self-reported, not objectively measured (no actigraphy or performance tests)
Sample characteristics	Young skew (median age 20); Female skew (60%); Mix of employees and students; May not generalize to older or male-dominated populations
Cross-sectional design	Cannot establish causality (does poor sleep cause low productivity, or vice versa?)
No clinical validation	The indexes have not been formally validated against clinical sleep assessments
Single time point	Captures only one moment; sleep patterns and productivity vary over time

6.3 Practical Implications

6.3.1 For Individuals (Employees and Students)

- Prioritize 7-9 hours of sleep
- Reduce device use 60 minutes before bed
- Maintain consistent sleep schedule (even on weekends)
- Address cognitive symptoms (poor concentration, memory lapses) as early warning signs

6.3.2 For Employers and Educational Institutions

- Implement sleep education programs
- Consider flexible start times to accommodate chronotypes
- Discourage after-hours email and late-night assignments
- Create rest areas for power naps (10-20 minutes)
- Address workload and stress as they interact with sleep

6.3.3 For Future Research

- Objective sleep measurement (actigraphy)
- Longitudinal design to establish causality
- Separate analysis for employees vs. students
- Diverse sample (older workers, more males, various industries and academic levels)
- Intervention studies (testing sleep improvement programs)

7 Conclusion

This study investigated the impact of sleep on work and academic productivity among 127 respondents (a mix of employees and students) using a 20-item questionnaire measuring sleep habits, cognitive performance, emotional regulation, physical safety, and presenteeism. From these data, five composite indexes were constructed: Sleep Quality & Hygiene Index (SQLI), Cognitive Task Performance Index (CTPI), Emotional & Social Impact Index (ESII), Physical & Safety Risk Index (PSRI), Presenteeism Index (PI), and an overall Sleep-Productivity Impact Index (SPI).

Descriptive analysis revealed that 65% of respondents obtain less than the recommended 7 hours of sleep per night. The mean SQLI score was 68.2 (SD = 11.8), indicating poor sleep quality. The mean CTPI was 56.6 (SD = 16.0), indicating moderate cognitive impairment. The mean ESII was 57.5 (SD = 16.6), indicating moderate emotional impact. The mean PSRI was 56.5 (SD = 18.2), indicating moderate safety risk. The mean PI was 20.7 (SD = 16.0), indicating low to moderate presenteeism. The overall SPI mean was 57.4 (SD = 14.6), ranging from 26 to 103. Respondents reported an average of 10.3 unproductive days per month due to poor sleep.

Bivariate correlation analysis revealed several notable relationships. Age showed a moderate negative correlation with SPI (r = -0.50), indicating that younger individuals experience greater sleep-related productivity impact. The CTPI demonstrated a strong positive correlation with SPI (r = +0.79), representing the strongest relationship in the study. The ESII showed a strong positive correlation with SPI (r = +0.73), and the PSRI showed a strong positive correlation with SPI (r = +0.74). The PI showed a moderate positive correlation with SPI (r = +0.66). In contrast, the SQLI showed only a weak positive correlation with SPI (r = +0.20). The ESII and CTPI were moderately correlated (r = +0.42), indicating that emotional dysregulation and cognitive difficulties tend to co-occur.

Inferential statistical testing yielded the following results. An independent samples t-test comparing males (n = 47) and females (n = 79) on SPI scores produced a t-statistic of -3.73 (df = 124) with a p-value of 0.00029. Since the p-value is below 0.05, the null hypothesis of no gender difference was rejected, indicating a statistically significant difference with females showing higher (worse) SPI scores.

A one-way ANOVA comparing SPI scores across five sleep duration categories (less than 5 hours, 5-6 hours, 6-7 hours, 7-8 hours, more than 8 hours) produced an F-statistic of 0.7509 with a p-value of 0.5242. Since the p-value exceeds 0.05, the null hypothesis of equal means across all groups was not rejected, indicating no statistically significant difference in SPI scores based on sleep duration alone.

Pearson correlation tests revealed statistically significant positive linear relationships for all examined pairs. The correlation between CTPI and SPI was r = 0.7946 (p ¡ 0.0001), with a coefficient of determination of r^2 = 0.63, indicating that approximately 63% of the variance in SPI scores is explained by cognitive performance. The correlation between PSRI and SPI was r = 0.7444 (p ¡ 0.0001), with r^2 = 0.55. The correlation between ESII and CTPI was r = 0.4156 (p ¡ 0.0001). The correlation between SQLI and SPI was r = 0.1974 (p = 0.0262).

In summary, the statistical evidence supports that cognitive performance has the strongest association with overall productivity impact, explaining 63% of the variance in SPI scores. Physical safety risk and emotional regulation also show strong associations with productivity impact. A statistically significant gender difference exists, with females showing higher impact scores. Sleep duration alone does not demonstrate a statistically significant association with productivity impact. Perceived sleep quality shows only a weak statistically significant association with productivity impact. Younger individuals experience greater sleep-related productivity loss, as indicated by the moderate negative correlation between age and SPI (r = -0.50). These findings provide statistical evidence that sleep-related productivity impairment is multidimensional, with cognitive performance being the primary driver, and that demographic factors such as gender and age moderate this relationship. The non-significant ANOVA results for sleep duration suggest that sleep quantity alone may be less predictive of productivity impact than sleep quality and consistency.

Appendix: Questionnaire

Part A: Sleep Habits & Quality (Questions 1-6)

Q1. On a typical work or school night, how many hours of sleep do you get?

- Less than 5 hours
- 5-6 hours
- 6-7 hours
- 7-8 hours
- More than 8 hours

Q2-Q6: 5-point Likert scale (1=Never to 5=Always)

8. I wake up feeling refreshed and ready for my day.
9. I have difficulty falling asleep the night before an important work or study day.
10. I wake up multiple times during the night and struggle to go back to sleep.
11. On weekends or days off, my sleep schedule is very different from my work/school day schedule.
12. I use electronic devices in bed within 30 minutes of trying to sleep.

Part B: Cognitive & Task Performance (Questions 7-12)

5-point Likert scale (1=Never to 5=Always)

13. I find it hard to concentrate on tasks for more than 15-20 minutes at a time.
14. I make careless mistakes that I would not have made if better rested.
15. I often have to re-read emails, documents, or instructions because I lost focus.
16. Learning a new software, process, or skill feels unusually difficult for me.
17. I struggle to generate new ideas or creative solutions to problems.
18. I forget important tasks, deadlines, or details from meetings or classes.

Part C: Emotional & Social Impact (Questions 13-16)

5-point Likert scale (1=Never to 5=Always)

19. I feel irritable or short-tempered with colleagues, classmates, or clients.
20. Small setbacks or minor criticism feel overwhelming or upsetting.
21. I avoid collaboration or group work because I feel mentally exhausted.
22. I have difficulty controlling my emotions during stressful situations (e.g., tight deadlines, difficult customers, exams).

Part D: Physical & Safety Factors (Questions 17-18)

5-point Likert scale (1=Never to 5=Always)

23. I feel physically tired, sluggish, or heavy-eyed during the day, even without heavy physical exertion.
24. I have had a close call, minor accident, or injury that I attribute to being too tired (e.g., tripping, bumping into things, spilling liquids).

Part E: Overall Productivity & Presenteeism (Questions 19-20)

Q19. Compared to when I am well-rested, I estimate my current output is...

- 80-100% less
- 60-80% less
- 40-60% less
- 20-40% less
- About the same or better

Q20. In the past month, how many days did you feel you were ”present but not productive” due to poor sleep?

- ___days (0-30)